METRI ENGINEERING SERVICES W.L.L.

TECHNICAL & COMMERCIAL PROPOSAL

HVAC PACKAGE FOR SIEMENS E-HOUSE

FEED FOR DESALINATION PLANT FOR NGL-5

MESAIEED INDUSTRIAL CITY, MDSP

Client: DANEM ENGINEERING WORKS W.L.L

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1. EXECUTIVE SUMMARY

Metri Engineering Services W.L.L. (MES) is pleased to submit this Technical and Commercial Proposal for the complete HVAC Package for the Siemens E-House at the NGL-5 Desalination Plant, Mesaieed Industrial City (MDSP). This proposal is prepared in full compliance with the FEED documents issued by Black Cat Consulting & Engineering Services W.L.L., approved for detailed design (Rev. 2, dated 30/09/2024).

1.1 Project Overview

The E-House HVAC system serves critical electrical infrastructure including the Switchboard Room, LER Room, UPS Room, Battery Room, Operator Room, and Airlock. The system is designed for continuous 24/7 operation with N+1 redundancy to ensure uninterrupted cooling and ventilation for sensitive electrical equipment.

1.2 Key Design Parameters

Parameter	Value
Design Ambient Condition	47°C DB / 30°C WB
Maximum Operating Ambient	52°C (equipment derate acceptable)
Indoor Design Temp (Summer)	23-24°C DB
Indoor Design Temp (Winter)	20°C DB (with electric reheat)
Relative Humidity Range	40% - 60% RH
Building Pressurization	+50 Pa (positive)
Total Cooling Capacity (Zone 1)	186.1 kW (52.9 TR)
Total Cooling Capacity (Zone 2)	412.2 kW (117.2 TR)

2. PROJECT UNDERSTANDING & ANALYSIS

2.1 E-House Building Areas

The E-House comprises the following functional areas requiring HVAC services:

Area
Battery Room
LER Room
Operator Room
Airlock
UPS Room
Switchboard Room

2.2 System Architecture

The HVAC system is designed with two independent zones for enhanced reliability:

- PACU-01 System (Zone 1): Serves LER Room, UPS Room, Battery Room, Operator Room, and Airlock with 2×100% redundancy configuration (1 Duty + 1 Standby)
- PACU-02 System (Zone 2): Serves Switchboard Room exclusively with 3×50% redundancy configuration (2 Duty + 1 Standby)
- Battery Room Extract System: ATEX-rated exhaust fans for hydrogen gas ventilation (Zone 1, IIC, T3)
- Purge System: Post fire-suppression gas purging with dedicated exhaust and make-up air fans

3. TECHNICAL SCOPE OF WORK

3.1 HVAC Design Engineering

Complete HVAC detailed design and engineering services including:

- Detailed cooling load calculations verification using Carrier HAP software
- Ductwork sizing and pressure drop calculations per SMACNA standards
- Equipment selection and vendor coordination
- Detailed installation drawings and shop drawings
- HVAC control system design and sequences of operation
- Coordination with electrical, instrumentation, and civil disciplines

3.2 Equipment Supply

Supply of all HVAC equipment as per the equipment schedules including:

- Packaged Air Conditioning Units (PACUs) with hermetic scroll compressors
- ATEX-rated Battery Room Extract Fans (Zone 1, IIC, T3)
- Purge Exhaust Fans and Make-up Air Fans
- Duct Mounted Electric Heaters (UL Listed)
- Complete ductwork system (GI, pre-insulated where required)
- Gas tight dampers, motorized dampers, and volume control dampers
- Supply and return air grilles, diffusers
- HVAC controls panel with complete instrumentation

3.3 Installation Works

- Complete installation of all HVAC equipment on prepared foundations/plinths
- Ductwork fabrication and installation with proper supports
- Condensate drainage piping
- Thermal insulation with Heresite/Blygold coating on coils
- Electrical connections to equipment (from MCC terminals).
- Control wiring and instrumentation installation

4. BILL OF QUANTITIES

4.1 Packaged Air Conditioning Units (PACUs)

S.No	Tag No.	Description / Location	Qty	Capacity (kW)	Airflow (L/s)
1	424-45-HAC-004	PACU Zone 1 - LER, UPS, Battery Room (Duty)	1	186.1	6,311
2	424-45-HAC-005	PACU Zone 1 - LER, UPS, Battery Room (Standby)	1	186.1	6,311
3	424-45-HAC-006	PACU Zone 2 - Switchboard Room (Duty 1)	1	206.1	13,465
4	424-45-HAC-007	PACU Zone 2 - Switchboard Room (Duty 2)	1	206.1	13,465
5	424-45-HAC-008	PACU Zone 2 - Switchboard Room (Standby)	1	206.1	13,465
		TOTAL PACUs	**5**		

4.2 Battery Room Extract Air Fans (ATEX Rated)

S.No	Tag No.	Description	Qty	Airflow (L/s)	ESP (Pa)
1	424-45-HEF-001	Battery Room Extract Fan - Duty (Zone 1, IIC, T3, Ex-de)	1	998	300
2	424-45-HEF-002	Battery Room Extract Fan - Standby (Zone 1, IIC, T3, Ex-de)	1	998	300
		TOTAL Battery Extract Fans	**2**		

4.3 Purge System Fans

S.No	Tag No.	Description	Qty	Airflow (L/s)	ESP (Pa)
1	424-45-HEF-003	Purge Exhaust Fan - Floor Mounted Ducted Axial	1	931	300
2	424-45-HMF-001	Make-up Air Fan - Floor Mounted Ducted Axial	1	931	300
		TOTAL Purge System Fans	**2**		

4.4 Duct Mounted Electric Heaters

S.No	Tag No.	Description / Location	Qty	Size (W×H mm)	Capacity (kW)
1	424-45-HEH-001	Electric Duct Heater - LER & UPS Room	1	1200 × 650	9.0
2	424-45-HEH-002/003	Electric Duct Heater - Switchboard Room	2	1650 × 1000	10.0
		TOTAL Electric Heaters	**3**		**29.0**

4.5 Dampers & Accessories

S.No	Description	Qty	Remarks
1	Gas Tight Dampers (External wall crossings)	As Reqd	Per design
2	Motorized Isolation Dampers	As Reqd	Per design
3	Volume Control Dampers (VCD)	As Reqd	Per design
4	Flexible Connectors	As Reqd	All equipment
5	Anti-Vibration Mounts / Support Feet	As Reqd	All rotating equipment
6	Disconnect Switches (Local Isolators)	As Reqd	All motors

4.6 BOQ Summary - Major Equipment

S.No	Equipment Description	Quantity	Unit
1	Packaged Air Conditioning Units (PACU) - Complete with DX coils, compressors, fans, filters, controls	5	Nos
2	Battery Room Extract Fans - ATEX Rated (Zone 1, IIC, T3, Ex-de)	2	Nos
3	Purge Exhaust Fan - Floor Mounted Ducted Axial	1	Nos
4	Make-up Air Fan - Floor Mounted Ducted Axial Supply	1	Nos
5	Duct Mounted Electric Heaters - UL Listed, Tubular Finned Type	3	Nos
6	HVAC Main Control Panel (Switchboard Room)	1	Lot
7	Purge Fan Control Panel (Airlock Room)	1	Lot
8	Complete Ductwork System (Supply, Return, Fresh Air, Exhaust) - GI with insulation	1	Lot
9	Grilles, Diffusers, Louvers - Supply, Return, Fresh Air	1	Lot
10	Dampers (Gas Tight, Motorized, VCD)	1	Lot
11	Refrigerant Piping & Condensate Drainage (complete)	1	Lot
12	Thermal Insulation (Ductwork & Piping)	1	Lot

5. TECHNICAL SPECIFICATIONS COMPLIANCE

5.1 PACU Technical Requirements

Parameter	Specification Compliance
Compressor Type	Hermetic Scroll - COMPLIANT
Refrigerant	R-134a / R-407C / R-410A - COMPLIANT
Evaporator Fan Type	Centrifugal with EC Motors - COMPLIANT
Condenser Fan Type	Axial Type - COMPLIANT
Pre-Filter	ePM10 <65% (EN ISO 16890) - COMPLIANT
Bag Filter	ePM1 ≥80% (EN ISO 16890) - COMPLIANT
Coil Coating	Heresite / Blygold or equivalent - COMPLIANT
Coil Face Velocity	Max 2.5 m/s (with moisture eliminator if exceeded) - COMPLIANT
Air Flow Direction	Top Discharge / Side Return - COMPLIANT
Operating Temperature	Up to 52°C ambient without failure - COMPLIANT
Area Classification	Non-Hazardous Area - COMPLIANT

5.2 Battery Room Extract Fans - ATEX Requirements

Parameter	Specification Compliance
Hazardous Area Classification	Zone 1 - COMPLIANT
ATEX Rating	Ex-de (Flameproof/Increased Safety) - COMPLIANT
Gas Group	IIC (Hydrogen) - COMPLIANT
Temperature Class	T3 (200°C max surface temp) - COMPLIANT
Fan Type	Ducted Inline Axial or Box Type Centrifugal - COMPLIANT
Ventilation Rate	12 ACH minimum - COMPLIANT (998 L/s)

6. SCOPE INCLUSIONS & EXCLUSIONS

6.1 Scope Inclusions

The following items and services are INCLUDED in this proposal:

6.1.1 Design & Engineering

- Complete HVAC detailed design and engineering
- Cooling load calculations verification (Carrier HAP)
- Ductwork sizing and layout drawings
- Equipment shop drawings and submittals
- Control system design and sequences
- As-built documentation

6.1.2 Equipment Supply

- 5 × Packaged Air Conditioning Units (complete)
- 2 × ATEX-rated Battery Room Extract Fans
- 1 × Purge Exhaust Fan
- 1 × Make-up Air Fan
- 3 × Duct Mounted Electric Heaters (UL Listed)
- Complete ductwork system (fabrication & supply)
- All dampers (gas tight, motorized, VCD)
- Grilles, diffusers, and louvers
- HVAC Control Panel and Purge Control Panel
- All instrumentation (temperature, pressure, humidity sensors)
- Flexible connectors and anti-vibration mounts
- Local disconnect switches

6.1.3 Installation Works

- Equipment installation on prepared foundations
- Ductwork installation with supports
- Refrigerant piping installation
- Condensate drainage installation
- Thermal insulation installation
- Control wiring and instrumentation
- Equipment electrical connections (from MCC terminals)

6.1.4 Testing & Commissioning

- Pre-commissioning checks
- Ductwork leak testing
- Equipment functional testing
- TAB (Testing, Adjusting, Balancing)
- Performance verification
- Documentation and handover

6.2 Scope Exclusions

The following items are EXCLUDED from this proposal:

- Civil works (foundations, plinths, structural steel)
- Building envelope penetrations (wall/roof openings)
- Power supply to MCC (by Electrical Contractor)
- Isolators, MCC panels and power cables to equipment
- DCS/ESD integration and programming
- Fire & Gas system integration (interface only)
- Clean agent fire suppression system
- Painting and finishing (beyond factory finish)
- Spare parts (quoted separately if required)
- Extended warranty beyond standard terms
- Operation and maintenance manuals (O&M) beyond standard documentation

7. COMMERCIAL TERMS & CONDITIONS

Lump sum price= 2,796,428 Q.R.

7.1 Payment Terms

Milestone	Description	Payment (%)
Advance	Upon Contract Award / PO Issuance	30%
Milestone 1	Design Approval & Material Procurement	25%
Milestone 2	Equipment Delivery to Site	20%
Milestone 3	Installation Completion	15%
Final	Testing, Commissioning & Handover	10%

7.2 Delivery Schedule

Activity	Duration (from Contract Award)
Design & Engineering Submittals	6 Weeks
Equipment Manufacturing & Delivery	20 Weeks
Installation & Commissioning	12 Weeks (after delivery)
Total Project Duration	**38 Weeks**

7.3 Warranty

- Equipment Warranty: 18 months from delivery or 12 months from commissioning, whichever occurs first
- Installation Workmanship Warranty: 12 months from commissioning
- Compressor Warranty: 12 months

7.4 Validity

This proposal is valid for 90 days from the date of submission. Prices are subject to revision after validity period.

7.5 Country of Origin

Major equipment will be sourced from reputable manufacturers with proven track record in similar QatarEnergy projects. Alternative origins may be proposed subject to QatarEnergy approval.

Best Regards,

METRI ENGINEERING SERVICES W.L.L.

Eng. Fayez Metri

متري للخدمات الهندسية ذ.م.م
ص.ب: ١٩٩٦٢، الدوحة - قطر
تليفاكس: ٤٤١٢٧٦٧
P.O. Box 19962, Doha-Qatar
TelFax: 4412767
METRI ENGINEERING SERVICES W.L.L